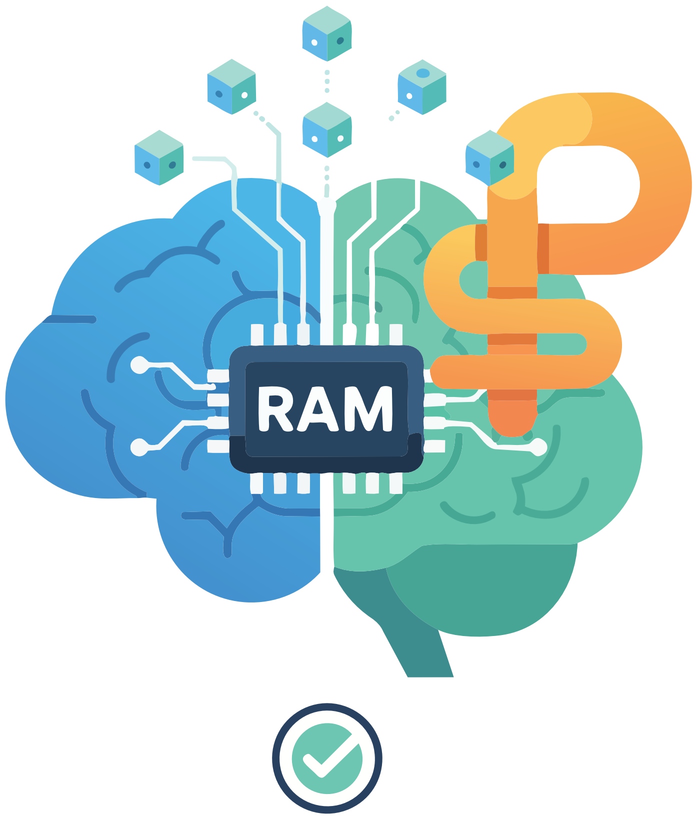
Optimising RAM Usage with Python

Discover how we can make better use of RAM by applying various Python optimisation techniques.



In today's world, with the abundance of RAM available, we rarely think about optimising our code. But sooner or later, and before we can realise it, we hit the limits of the hardware.

Let's begin with how to measure the amount of RAM used by a Python program. There is always more than one way to do so and each has its merits.

Using *sys.getsizeof*

This is a built-in function from the sys module that returns the shallow size of a single Python object in bytes.

What it measures: It measures only the memory directly attributed to the object itself (e.g., the list structure), excluding the size of any referenced objects (e.g., the integers or strings inside a list).

The pros are:

- Simple, fast and lightweight.
- Useful for quick comparisons of basic object overhead (e.g., empty list vs small list).

Cons/limitations:

- Underestimates containers like lists, dicts, or custom objects with reference values in them.
- As it is not recursive, it doesn't help with complex structures or total memory footprint.

When to use: Use it for basic, one-off checks on simple objects or understanding Python's object overhead. It's not suitable for profiling real memory usage in applications.

```
import sys

def get_object_size_mb(obj):
    """
    Returns the size of a Python object in megabytes (MB).
    """
    size_bytes = sys.getsizeof(obj)
    size_mb = size_bytes / (1024 * 1024) # Convert bytes to
MB
    return size_mb

# create huge list
a = [i for i in range(5_000_000)]
size_of_list_mb = get_object_size_mb(a)
print(f"Size python object is: {size_of_list_mb:.2f} MB")

##OUTPUT
% /usr/bin/python3 getsizeof_sample.py Size python object is:
38.35 MB
```

Using *tracemalloc*

This built-in standard library module (since Python 3.4) traces Python memory allocations.

What it measures: It tracks memory blocks allocated by Python, providing detailed statistics (size, count, traceback) grouped by line, file, or traceback. You can take snapshots and